Case Studies in Operations Management

Mathematical Model Milestone 1

Business and Logistics Plan for "Fresh Veggie" (FV)

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1 Model Assumptions

The model corresponds to a first-stage strategic network design. It evaluates a set of predefined facility configurations and selects the one that maximizes estimated annual profit.

The following simplifying assumptions are adopted:

- Demand is constant over the year and derived from per-capita consumption.
- Each facility configuration $z \in \mathcal{Z}$ has a precomputed coverage level $p(z)$ measured in number of people.
- The company can serve at most a fraction α^{max} of the covered population.
- Revenue is estimated using per-capita annual demand and average selling prices.
- Only fixed opening costs of facilities are considered at this stage.
- The model includes revenues, sourcing costs, transportation costs, and fixed facility opening costs, but does not yet incorporate inventory dynamics, perishability, or service-time uncertainty.

2 Sets

- DC: set of candidate distribution centers.
- DS: set of candidate dark stores.
- $MDCP$: set of candidate parking locations for mobile distribution centers.
- $F = DC \cup DS \cup MDCP$: set of all candidate facilities in the network.
- P: set of ports:
$$P = \{\text{San Antonio}, \text{Valparaíso}\}$$
- A: set of airports:
$$A = \{\text{Arturo Merino Benítez International Airport}\}$$
- LC: Land customs:
$$LC = \{\text{Aduana Chacalluta}\}$$
- $E = P \cup A \cup LC$: set of international entry points to Chile.
- WM: set of wholesale markets:
$$WM = \{\text{La Vega Central}, \text{Lo Valledor}\}$$
- $\mathcal{Z}$: set of all feasible facility configurations, where each $z \in \mathcal{Z}$ is a binary vector indicating which facilities are active.
- S^{imp}: set of imported supplier countries:
$$S^{imp} = \{\text{US}, \text{Mexico}, \text{Peru}, \text{Ecuador}, \text{Bolivia}, \text{Argentina}, \text{Spain}\}$$
- S^{cl}: set of domestic supplier regions in Chile:
$$S^{cl} = \{\text{Chile-Scl}, \text{Chile-Vap}, \text{Chile-Maul}, \text{Chile-Coq}\}$$
- $S = S^{imp} \cup S^{cl}$: set of all supplier origins, where
$$S^{imp} \cap S^{cl} = \varnothing$$
- K: set of products:
$$K = \{\text{Roma Tomatoes}, \text{Cauliflower}, \text{Broccoli}, \text{Asparagus}, \text{Green Bell Pepper}\}$$
- M: set of months:
$$M = \{\text{Jan}, \text{Feb}, \text{Mar}, \text{Apr}, \text{May}, \text{Jun}, \text{Jul}, \text{Aug}, \text{Sep}, \text{Oct}, \text{Nov}, \text{Dec}\}$$

3 Parameters

3.1 Demand and Revenue Parameters

- $p(z)$: number of people covered by configuration $z \in \mathcal{Z}$.
- α^{max}: maximum fraction of covered demand that can be served.
- $d^{pc}_{k,m}$: per-capita demand of product $k \in K$ in month $m \in M$, expressed in kg/person/month.
- P_k: selling price per kilogram of product $k \in K$, expressed in \$/kg.

3.2 Facility Parameters

- $b_{f,z}$: binary parameter equal to 1 if facility $f \in F$ is included in configuration z, and 0 otherwise.
- C^{open}_f: fixed cost of opening facility $f \in F$, expressed in \$.

3.3 Availability Parameters

- $a_{k,s,m}$: binary parameter equal to 1 if product $k \in K$ can be sourced from supplier $s \in S$ during month $m \in M$, and 0 otherwise.
- p^F_i: number of people covered by facility $i \in F$.

3.4 Sourcing Costs

- $C^{pur,imp}_{k,s,e}$: purchasing cost per kilogram of product $k \in K$ acquired from imported supplier $s \in S^{imp}$ through international entry point $e \in E$, expressed in \$/kg.
- $C^{pur,cl}_{k,s}$: purchasing cost per kilogram of product $k \in K$ acquired from domestic supplier $s \in S^{cl}$, expressed in \$/kg.
- $C^{pur,wm}_{k,w}$: purchasing cost per kilogram of product $k \in K$ acquired from wholesale market $w \in WM$, expressed in \$/kg.

3.5 Distance Parameters

- $D^{DC,DS}_{i,j}$: distance between distribution center $i \in DC$ and dark store $j \in DS$, expressed in km.
- $D^{DC,MDCP}_{i,n}$: distance between distribution center $i \in DC$ and mobile distribution center parking location $n \in MDCP$, expressed in km.
- $D^{E,DC}_{e,i}$: distance between international entry point $e \in E$ and distribution center $i \in DC$, expressed in km.
- $D^{S^{cl},DC}_{s,i}$: distance between domestic supplier $s \in S^{cl}$ and distribution center $i \in DC$, expressed in km.
- $D^{WM,DC}_{w,i}$: distance between wholesale market $w \in WM$ and distribution center $i \in DC$, expressed in km.

- $D_{n,r}^{MDCP,MDCP}$: distance between mobile distribution center parking locations $n, r \in MDCP$, expressed in km.
- D_i^{lm}: average last-mile delivery distance from facility $i \in F$ to customers within its service area, expressed in km.
- D_i^{MDC}: minimum route distance traveled by the mobile distribution center when departing from distribution center $i \in DC$, visiting all assigned mobile distribution center parking locations, and returning to the same distribution center, expressed in km.

The average last-mile distance is approximated assuming that customers are uniformly distributed within the service area of each facility. Under this assumption, the expected distance from the facility to a customer is given by $\frac{2}{3}$ of the service radius.

$$D_i^{lm} = \begin{cases} \frac{2}{3} \cdot 15, & \text{if } i \in DC \\ \frac{2}{3} \cdot 5, & \text{if } i \in DS \\ \frac{2}{3} \cdot 10, & \text{if } i \in MDCP \end{cases}$$

3.6 Transportation Cost Parameters

- $c_k^{tr,long}$: long-haul transportation cost per kilogram-kilometer of product $k \in K$, expressed in \$/kg/km.
- $c_k^{tr,int}$: inter-facility transportation cost per kilogram-kilometer of product $k \in K$, expressed in \$/kg/km.
- c_k^{lm}: last-mile transportation cost per kilogram-kilometer of product $k \in K$, expressed in \$/kg/km.

3.7 MDC Transportation Cost Parameter

- c^{mov}: operating cost of moving the mobile distribution center per kilometer traveled, expressed in \$/km.

3.8 Auxiliary Capacity Parameters

- $M_{k,s,e,m}^{imp}$: sufficiently large upper bound on the quantity of imported product k that can be purchased from supplier s through entry point e in month m.
- $M_{k,s,m}^{cl}$: sufficiently large upper bound on the quantity of domestic product k that can be purchased from supplier s in month m.
- $M_{k,i,m}^{del}$: sufficiently large upper bound on the quantity of product k that can be delivered from facility i to final customers in month m.
- $M_{k,j,m}^{DS}$: sufficiently large upper bound on the quantity of product k that can be received by dark store j in month m.
- $M_{k,m}^{MDC}$: sufficiently large upper bound on the quantity of product $k \in K$ that can be loaded from a distribution center to the mobile distribution center during month $m \in M$.

4 Decision Variables

4.1 Configuration Selection

- $y_z \in \{0, 1\}$: binary variable equal to 1 if configuration $z \in \mathcal{Z}$ is selected, and 0 otherwise.

4.2 Sourcing Decisions

- $q^{imp}_{k,s,e,m} \geq 0$: quantity of product $k \in K$ purchased during month $m \in M$ from imported supplier $s \in S^{imp}$ through international entry point $e \in E$, expressed in kg.
- $q^{cl}_{k,s,m} \geq 0$: quantity of product $k \in K$ purchased during month $m \in M$ from domestic supplier $s \in S^{cl}$, expressed in kg.
- $q^{wm}_{k,w,m} \geq 0$: quantity of product $k \in K$ purchased during month $m \in M$ from wholesale market $w \in WM$, expressed in kg.

4.3 Flow Variables

- $x^{E,DC}_{k,e,i,m} \geq 0$: quantity of product $k \in K$ transported from entry point $e \in E$ to distribution center $i \in DC$ during month m, expressed in kg.
- $x^{S^{cl},DC}_{k,s,i,m} \geq 0$: quantity of product $k \in K$ transported from domestic supplier $s \in S^{cl}$ to distribution center $i \in DC$ during month m, expressed in kg.
- $x^{WM,DC}_{k,w,i,m} \geq 0$: quantity of product $k \in K$ transported from wholesale market $w \in WM$ to distribution center $i \in DC$ during month m, expressed in kg.
- $x^{DC,DS}_{k,i,j,m} \geq 0$: quantity of product $k \in K$ transported from distribution center $i \in DC$ to dark store $j \in DS$ during month m, expressed in kg.
- $x^{DC,MDC}_{k,i,m} \geq 0$: quantity of product $k \in K$ transported from distribution center $i \in DC$ to the mobile distribution center during month $m \in M$, expressed in kg.

4.4 MDC Assignment Variables

- $u_{j,m} \in \{0, 1\}$: binary variable equal to 1 if distribution center $j \in DC$ supplies the mobile distribution center during month $m \in M$, and 0 otherwise.

4.5 Delivery

- $x^{F,C}_{k,i,m} \geq 0$: quantity of product $k \in K$ delivered from facility $i \in F$ to final customers during month $m \in M$, expressed in kg.

5 Objective Function

The objective is to maximize total profit, defined as total sales revenue minus sourcing, transportation, and facility opening costs.

5.1 Revenue

Revenue is obtained from the quantity of product delivered to final customers.

$$\text{Revenue} = \sum_{m \in M} \sum_{k \in K} \sum_{i \in F} P_k \, x_{k,i,m}^{F,C}$$

5.2 Acquisition Cost

Acquisition cost includes purchases from imported suppliers, domestic suppliers, and wholesale markets.

$$\text{Acquisition Cost} = \sum_{m \in M} \sum_{k \in K} \left(\sum_{s \in S^{imp}} \sum_{e \in E} C_{k,s,e}^{pur,imp} \, q_{k,s,e,m}^{imp} + \sum_{s \in S^{cl}} C_{k,s}^{pur,cl} \, q_{k,s,m}^{cl} + \sum_{w \in WM} C_{k,w}^{pur,wm} \, q_{k,w,m}^{wm} \right)$$

5.3 Transportation Cost

Transportation cost is divided into long-haul transportation, inter-facility transportation, and last-mile delivery.

5.3.1 Long-Haul Transportation Cost

This cost includes product movements from international entry points, domestic suppliers, and wholesale markets to distribution centers.

$$\begin{aligned} \text{Long-Haul Transportation Cost} = \sum_{m \in M} \sum_{k \in K} \Bigg(& \sum_{e \in E} \sum_{i \in DC} c_k^{tr,long} \, D_{e,i}^{E,DC} \, x_{k,e,i,m}^{E,DC} \\ & + \sum_{s \in S^{cl}} \sum_{i \in DC} c_k^{tr,long} \, D_{s,i}^{S^{cl},DC} \, x_{k,s,i,m}^{S^{cl},DC} \\ & + \sum_{w \in WM} \sum_{i \in DC} c_k^{tr,long} \, D_{w,i}^{WM,DC} \, x_{k,w,i,m}^{WM,DC} \Bigg) \end{aligned}$$

5.3.2 Inter-Facility Transportation Cost

This cost includes product flows between facilities inside the distribution network.

$$\text{Inter-Facility Transportation Cost} = \sum_{m \in M} \sum_{k \in K} \left(\sum_{i \in DC} \sum_{j \in DS} c_k^{tr,int} \, D_{i,j}^{DC,DS} \, x_{k,i,j,m}^{DC,DS} \right)$$

5.3.3 Last-Mile Transportation Cost

This cost is proportional to the delivered quantity and the average last-mile distance associated with each facility.

$$\text{Last-Mile Transportation Cost} = \sum_{m \in M} \sum_{k \in K} \sum_{i \in F} c_k^{lm} \, D_i^{lm} \, x_{k,i,m}^{F,C}$$

5.4 Facility Opening Cost

Facility opening cost depends on the selected configuration.

$$\text{Facility Opening Cost} = \sum_{z \in \mathcal{Z}} \left(\sum_{f \in F} C_f^{open} \, b_{f,z} \right) y_z$$

5.4.1 Mobile Distribution Center Movement Cost

This cost accounts for the movement of the mobile distribution center from a distribution center, through its assigned parking locations, and back to the same distribution center.

$$\text{MDC Movement Cost} = \sum_{m \in M} \sum_{i \in DC} c^{mov} \, D_i^{MDC} \, u_{i,m}$$

5.5 Complete Objective Function

$$\begin{aligned} \max \quad & \text{Revenue} \\ & - \text{Acquisition Cost} \\ & - \text{Long-Haul Transportation Cost} \\ & - \text{Inter-Facility Transportation Cost} \\ & - \text{Last-Mile Transportation Cost} \\ & - \text{MDC Movement Cost} \\ & - \text{Facility Opening Cost} \end{aligned}$$

6 Constraints

6.1 Configuration Selection

$$\sum_{z \in \mathcal{Z}} y_z = 1$$

6.2 Availability Constraints

$$q^{imp}_{k,s,e,m} \leq M^{imp}_{k,s,e,m}\, a_{k,s,m}, \quad \forall k \in K,\ \forall s \in S^{imp},\ \forall e \in E,\ \forall m \in M$$

$$q^{cl}_{k,s,m} \leq M^{cl}_{k,s,m}\, a_{k,s,m}, \quad \forall k \in K,\ \forall s \in S^{cl},\ \forall m \in M$$

6.3 Purchase-Flow Consistency

$$\sum_{i \in DC} x^{E,DC}_{k,e,i,m} = \sum_{s \in S^{imp}} q^{imp}_{k,s,e,m}, \quad \forall k \in K,\ \forall e \in E,\ \forall m \in M$$

$$\sum_{i \in DC} x^{S^{cl},DC}_{k,s,i,m} = q^{cl}_{k,s,m}, \quad \forall k \in K,\ \forall s \in S^{cl},\ \forall m \in M$$

$$\sum_{i \in DC} x^{WM,DC}_{k,w,i,m} = q^{wm}_{k,w,m}, \quad \forall k \in K,\ \forall w \in WM,\ \forall m \in M$$

6.4 Flow Balance at Distribution Centers

$$\begin{aligned}
&\sum_{e \in E} x^{E,DC}_{k,e,i,m} + \sum_{s \in S^{cl}} x^{S^{cl},DC}_{k,s,i,m} + \sum_{w \in WM} x^{WM,DC}_{k,w,i,m} \\
&= \sum_{j \in DS} x^{DC,DS}_{k,i,j,m} + x^{DC,MDC}_{k,i,m} + x^{F,C}_{k,i,m}, \quad \forall k \in K,\ \forall i \in DC,\ \forall m \in M
\end{aligned}$$

6.5 Activation Constraints

$$x^{F,C}_{k,i,m} \leq M^{del}_{k,i,m} \sum_{z \in \mathcal{Z}} b_{i,z} y_z, \quad \forall k \in K,\ \forall i \in F,\ \forall m \in M$$

$$\sum_{i \in DC} x^{DC,DS}_{k,i,j,m} \leq M^{DS}_{k,j,m} \sum_{z \in \mathcal{Z}} b_{j,z} y_z, \quad \forall k \in K,\ \forall j \in DS,\ \forall m \in M$$

6.6 Demand Constraints

Network-Level Demand Constraint

$$\sum_{i \in F} x^{F,C}_{k,i,m} \leq \alpha^{\max} \sum_{z \in \mathcal{Z}} p(z)\, d^{pc}_{k,m}\, y_z, \quad \forall k \in K,\ \forall m \in M$$

Facility-Level Demand Constraint

$$x^{F,C}_{k,i,m} \leq \alpha^{\max}\, d^{pc}_{k,m}\, p^F_i \sum_{z \in \mathcal{Z}} b_{i,z}\, y_z, \quad \forall k \in K,\ \forall i \in F,\ \forall m \in M$$

6.7 Mobile Distribution Center Constraints

$$\sum_{i \in DC} x_{k,i,m}^{DC,MDC} = \sum_{n \in MDCP} x_{k,n,m}^{F,C}, \quad \forall k \in K,\ \forall m \in M$$

$$\sum_{j \in DC} u_{j,m} = 1, \quad \forall m \in M$$

$$x_{k,j,m}^{DC,MDC} \leq M_{k,m}^{MDC}\, u_{j,m}, \quad \forall k \in K,\ \forall j \in DC,\ \forall m \in M$$

6.8 Binary Variables

$$y_z \in \{0,1\}, \quad \forall z \in \mathcal{Z}$$

$$u_{j,m} \in \{0,1\}, \quad \forall j \in DC,\ \forall m \in M$$

6.9 Non-Negativity Constraints

$$q_{k,s,e,m}^{imp},\ q_{k,s,m}^{cl},\ q_{k,w,m}^{wm} \geq 0, \quad \forall k \in K,\ \forall m \in M$$

$$x_{k,e,i,m}^{E,DC},\ x_{k,s,i,m}^{S^{cl},DC},\ x_{k,w,i,m}^{WM,DC},\ x_{k,i,j,m}^{DC,DS},\ x_{k,i,m}^{DC,MDC},\ x_{k,i,m}^{F,C} \geq 0$$